ANIMESH JAIN

PhD Student

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Fairfax, Virginia

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Scientific Computing: Numerical methods for PDEs, finite element methods, numerical linear algebra

Tools: LaTeX, Linux-based workflow

EDUCATION

01/2025 -(Present) -**PhD Mathematics** — **George Mason University, Fairfax, VA, USA**
Advisor: Dr. Harbir Antil

07/2022 -05/2024 -**M.Sc. Mathematics** — **Indian Institute of technology, Roorkee**
Overall CGPA- 8.66/10

07/2019 -05/2022 -**B.Sc. (Honours) Physics** — **Ramjas College, Delhi University**
Overall CGPA- 7.9/10

RESEARCH INTEREST

Numerical analysis and scientific computing for deterministic and stochastic partial differential equations, including finite element methods, PDE-constrained optimization, and operator-theoretic foundations, with applications to mathematical physics and related applied problems.

RESEARCH EXPERIENCE

Spring 2026 **Graduate Research Assistant** — **George Mason University** — **Advisor: Dr. Harbir Antil**

- Advancing theoretical and numerical aspects of PDE-constrained optimization.
- Further development and analysis of finite element solvers and adjoint-based optimization methods.
- Ongoing work on mathematical foundations and computational strategies for deterministic and stochastic PDE models.

Spring 2025 **Graduate Research Assistant** — **George Mason University** — **Advisor: Dr. Harbir Antil**

- Implemented finite element discretizations and developed numerical solvers for partial differential equations.
- Worked on optimization problems constrained by PDEs, focusing on adjoint-based methods for gradient computation.
- Studied foundational concepts in nonlinear functional analysis relevant to optimization with PDE constraints.

Spring 2024 **Master's Thesis: Deterministic and Stochastic Conservation Laws** — **Indian Institute of Technology, Roorkee** — **Supervisor: Prof. Ankik Kumar Giri**

- Analyzed first-order hyperbolic conservation laws, focusing on shock formation and global weak solutions.
- Applied Rankine–Hugoniot conditions and entropy methods.
- Extended the analysis to stochastic conservation laws with additive noise.

Fall 2023 **Reading Project: Functional Analysis and Applications** **Indian Institute of Technology, Roorkee** **Supervisor: Prof. Ankik Kumar Giri**

- Studied Sobolev spaces, weak convergence, compactness, and operator theory relevant to PDEs.

AWARDS AND HONOURS

Mar 2022 **IIT–JAM (Mathematics) — All India Rank 191**
(*National-level graduate entrance examination; highly selective with an acceptance rate of 6%.*)

Jul 2019 **Academic Excellence Award** **Ryan International School**

- Gold Medal for Rank-1 in Science Stream (Board Examinations)
- 7th Rank in the District

PUBLICATIONS

2026 **Fisher-Information-Based Sensor Placement for Structural Digital Twins: Analytic Results and Benchmarks** **with H. Antil, R. Löhner**
arXiv:2602.02981

- Developed a Fisher-information-based framework for optimal sensor placement in adjoint-based finite element models.
- Derived matrix-free operator formulations for efficient computation of Fisher information using forward and adjoint solves.
- Established analytical results for D-optimal sensor configurations, including uniform spacing in a 1D structural model.

2026 (Ongoing) **Conditional Value-at-Risk Regularization for Sparse and Measure-Concentrated Optimization** **with H. Antil, B. Mendieta**

- Developing CVaR-based regularization techniques to promote sparsity and measure concentration in optimization problems.
- Analyzing the formulation using tools from convex and nonlinear functional analysis, including dual representations and optimality conditions.
- Implementing and testing numerical methods for PDE-constrained optimization problems involving CVaR regularization.

SELECTED COURSEWORK

Functional Analysis I-II, Nonlinear Functional Analysis, Numerical Analysis, Theory of Partial Differential Equations, Measure Theory, Operator Theory, Financial Mathematics, Advanced Complex Analysis, Topology.

TEACHING EXPERIENCE

Fall 2025 **Graduate Teaching Assistant** **Ordinary Differential Equations, George Mason University**

- Led weekly recitation sessions focused on problem-solving and conceptual reinforcement.
- Supported students through office hours and guided practice on core ODE techniques.
- Assisted with grading homework/quizzes/exams and maintaining course logistics.

Fall 2025 **Seminar Leader** **PDE Control Seminar, George Mason University**

- Led seminar sessions for most of the semester on PDE theory through well-posedness of elliptic equations.
- Delivered additional seminar lectures on topics in nonlinear analysis relevant to PDE-constrained problems.

CONFERENCES & WORKSHOPS

Oct 17–18, 2025 **Finite Element Circus** **George Mason University, Arlington Campus**

- Attended invited talks and research presentations on finite element methods and numerical analysis of PDEs.
- Engaged with researchers and graduate students on current topics in computational PDEs.

REFERENCES

Prof. Harbir Antil hantil@gmu.edu
(PhD Supervisor)
George Mason Univeristy, Fairfax, VA.

Prof: Ankik Kumar Giri ankik.giri@ma.iitr.ac.in
(Master Thesis Supervisor)
Indian Institute of Technology Roorkee

Prof: Chaman Kumar chaman.kumar@ma.iitr.ac.in
(Department of Mathematics)
Indian Institute of Technology Roorkee